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Polypeptide dimer with high sialic acid content, comprising extracellular domain of alpha subunit of IGE FC receptor, and pharmaceutical composition comprising same

Abstract

A modified IgE Fc receptor having a high content of sialic acid and a pharmaceutical composition contains the modified IgE Fc receptor are disclosed. The polypeptide dimer having a high content of sialic acid not only has excellent safety and persistence in a body as compared with conventionally used anti-IgE antibodies, but also shows strong binding to IgE. Thus, the polypeptide has the advantage of an extended administration cycle. The polypeptide dimer is also an IgE single target substance, and unlike conventional anti-IgE antibodies to which the Fc of IgG1 is applied, does not bind to an Fc gamma receptor. The polypeptide dimer can be usefully used for the prevention or treatment of allergic diseases.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/KR2020/008855 filed Jul. 7, 2020, claiming priority based on Korean Patent Application No. 10-2019-0082217 filed Jul. 8, 2019.

SEQUENCE LISTING

(2) The content of the electronically submitted sequence listing, file name: Sequence_Listing_As_Filed.txt; file size: 36,504 bytes, date of creation: Jan. 1, 2022, filed herewith, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(3) The present invention relates to a polypeptide dimer comprising an IgE Fc receptor and having a high content of sialic acid and a pharmaceutical composition comprising the same.

BACKGROUND ART

(4) As the modern society is industrialized and eating habits are westernized, the incidence rate of allergic diseases, such as allergic rhinitis, atopic dermatitis, and food allergy, including asthma is increasing. Development of anaphylaxis, a severe allergic disease, is also increasing. These chronic immune diseases severely impair individuals' quality of life and socioeconomic costs are soaring accordingly. Thus, there is a desperate need for measures to overcome such diseases.

(5) Most allergic diseases are caused by an excessive immune response of immunoglobulin E (IgE). IgE is an antibody that is present in blood at a very low concentration under a normal condition. IgE is also usually produced by innocuous antigens. There is a case where the number of IgE is increased without any particular stimulus. Such a case may lead to allergic diseases. The abnormally increased number of IgE can bind to high-affinity IgE Fc receptors (FcεR1s) which are expressed on the surface of mast cells, basophils, and the like. Such binding causes mast cells or basophils to release chemical mediators such as histamine, leukotriene, prostaglandin, bradykinin, and platelet-activating factors. Release of these chemical mediators results in allergic symptoms. In particular, allergic diseases may exhibit worsened symptoms due to the binding between IgE and FcεR1. In addition, FcεR1-expressing cells are known to increase in allergic patients.

(6) Currently, various methods, such as allergen avoidance, administration of anti-allergic drugs, modulation of IgE synthesis in the body, and development of anti-IgE antibodies, have been proposed to treat allergic diseases. However, there still remain many drawbacks, such as insufficient drug efficacy and occurrence of serious side effects. Recently, a composition capable of binding to IgE and FcγRIIb with high affinity and inhibiting cells that express membrane-anchored IgE has been studied. This composition has been reported to be useful for treating IgE-mediated diseases including allergy and asthma (KR10-1783272B).

(7) In particular, omalizumab (trade name: Xolair), which targets an Fc portion of an IgE antibody, has been developed and used as a therapeutic agent for intractable severe asthma and intractable urticaria. However, a high-dose administration of omalizumab to maintain therapeutic effects leads to a high cost burden, and side effects such as angioedema and anaphylactic reaction (The Journal of Clinical Investigation Volume 99, Number 5, March 1997, 915-925).

(8) Besides, from the post-marketing results, serious adverse reactions such as allergic granulomatous vasculitis and idiopathic severe thrombocytopenia have been reported.

DETAILED DESCRIPTION OF THE INVENTION

Technical Problem

(9) Accordingly, the present inventors studied to develop a safe and effective therapeutic agent for allergic diseases. As a result, it was found that when the sialic acid content of a polypeptide dimer comprising two monomers (FcεRIα-ECD) comprising an extracellular domain of an alpha subunit of an IgE Fc receptor is high, the above polypeptide dimer has excellent binding affinity to IgE, as well as maintains a high concentration in the blood. In particular, the present invention was completed by confirming that even when the polypeptide dimer having a high content of sialic acid is administered subcutaneously, effective delivery into the body is possible.

Solution to Problem

(10) An aspect of the present invention provides a polypeptide dimer comprising two monomers comprising an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor, wherein the molar ratio of sialic acid/polypeptide dimer is at least 8.

(11) Another aspect of the present invention provides a pharmaceutical composition for preventing or treating allergic diseases, comprising the polypeptide dimer having a high content of sialic acid.

(12) Another aspect of the present invention provides a transdermal patch comprising the pharmaceutical composition.

(13) Another aspect of the present invention provides a topical patch comprising the pharmaceutical composition.

(14) Another aspect of the present invention provides a food composition for ameliorating or alleviating allergic symptoms, comprising the polypeptide dimer having a high content of sialic acid.

(15) Another aspect of the present invention provides a use of the polypeptide dimer having a high content of sialic acid for preventing or treating allergic diseases.

Effect of the Invention

(16) The polypeptide dimer having a high content of sialic acid according to the present invention not only has excellent safety and persistence in the body as compared with conventionally used anti-IgE antibodies, but also binds to IgE very strongly, which has the advantage of an extended administration cycle. In addition, the polypeptide dimer having a high content of sialic acid according to the present invention is an IgE single

target substance, and unlike conventional anti-IgE antibodies to which the Fc of IgG1 is applied, does not bind to an Fc gamma receptor. Thus, it can inhibit release of mediators caused by being bound to the Fc gamma receptor on the surface of mast cells, so that severe side effects such as occurrence of anaphylaxis which can be caused by binding between IgG1 and Fc gamma receptor III on mast cells can be minimized. In addition, the polypeptide dimer having a high content of sialic acid can maintain a high concentration in the blood even in the case of subcutaneous administration. Accordingly, the polypeptide dimer having a high content of sialic acid according to the present invention can be usefully used for the prevention or treatment of allergic diseases.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a view showing the results of confirming a polypeptide dimer produced in each cell line by SDS-PAGE.
- (2) FIG. 2 is a view showing SDS-PAGE results for a non-reduced form and a reduced form of a polypeptide dimer according to one embodiment.
- (3) FIG. 3 is a view showing a binding affinity of omalizumab to IgE.
- (4) FIG. 4 is a view showing a binding affinity of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention to IgE.
- (5) FIG. 5a is a view showing a binding affinity of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor I (FcγRI).
- (6) FIG. 5b is a view showing a binding affinity of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIA (FcγRIIA).
- (7) FIG. 5c is a view showing a binding affinity of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIB (FcγRIIB).
- (8) FIG. 5d is a view showing a binding affinity of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIIA (FcγRIIA).
- (9) FIG. 5e is a view showing a binding affinity of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention or omalizumab to IgG Fe gamma receptor IIIB (FcγRIIIB).
- (10) FIG. 6 is a graph obtained by quantifying a binding affinity between the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention or omalizumab and IgG Fc gamma receptors.
- (11) FIG. 7 is a view showing a degree of binding with IgE depending on the concentration of a polypeptide dimer according to an embodiment of the present invention.
- (12) FIG. 8 is a view showing a comparison between inhibitory abilities, on activity of human FcεRI-expressing mouse-derived mast cells, of the polypeptide dimer (IgE.sub.TRAP) according to an embodiment of the present invention and Xolair (omalizumab) depending on concentrations thereof.
- (13) FIG. 9 is a view confirming the anti-allergic effect of a polypeptide dimer according to an embodiment of the present invention by measuring the frequency of diarrhea in a food allergy model.
- (14) FIG. 10 is a view showing SDS-PAGE results for a non-reduced form and a reduced form of a polypeptide dimer according to an embodiment of the present invention (SA.sup.low, SA.sup.medi and SA.sup.high).
- (15) FIG. 11 is a view confirming the isoelectric point of a polypeptide dimer according to an embodiment of the present invention (SA.sup.low, SA.sup.medi and SA.sup.high).
- (16) FIG. 12 is a view confirming the anti-allergic effect of subcutaneous injection of a polypeptide dimer according to an embodiment of the present invention (SA.sup.low and SA.sup.high) by measuring the frequency of diarrhea in a food allergy model.
- (17) FIG. 13 is a view confirming the anti-allergic effect of a polypeptide dimer according to an embodiment of the present invention (SA.sup.low and SA.sup.high) by measuring the IgE concentration in the blood in a food allergy model.
- (18) FIG. 14 is a view confirming the anti-allergic effect of a polypeptide dimer according to one embodiment (SA.sup.low and SA.sup.high) by measuring the MCPT-1 concentration in the blood in a food allergy model.
- (19) FIG. 15 is a view showing the results of analysis of a pharmacokinetic profile of a polypeptide dimer according to one embodiment depending on sialic acid content in a mouse model.
- (20) FIG. 16 is a view confirming the anti-allergic effect of a polypeptide dimer according to one embodiment

depending on sialic acid content by measuring the body temperature in a passive systemic anaphylaxis mouse model.

DETAILED DESCRIPTION OF INVENTION

(21) An aspect of the present invention provides a polypeptide dimer having a high content of sialic acid, which comprises two monomers comprising an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor, wherein the monomer comprises an Fc region, and the Fc region and FcεRIα-ECD are linked via a hinge, and the polypeptide dimer is characterized in that the molar ratio of sialic acid/polypeptide dimer is at least 8.

(22) As used herein, the term “IgE” means an antibody known as immunoglobulin E. IgE has an affinity to mast cells, basophils, or the like. In addition, reaction between an IgE and an antigen (allergen) corresponding thereto causes an inflammatory reaction. In addition, IgE is known to be a major cause of anaphylaxis.

(23) As used herein, the term “IgE Fc receptor” is also referred to as Fcε receptor and indicates a receptor binding to an Fc portion of IgE. There are two types for the receptor. The receptor having high affinity to IgE Fc is called Fcε receptor I (FcεRI). The receptor having low affinity to IgE Fc is called Fcε receptor II (FcεRII). FcεRI is expressed in mast cells and basophils. In a case where IgE antibodies bound to FcεRI are cross-linked by polyvalent antigens, degranulation occurs in mast cells or basophils, thereby releasing various chemical transmitter substances including histamine. This release leads to an immediate allergic reaction.

(24) The FcεRI is a membrane protein composed of one α chain, one β chain, and two γ chains linked by a disulfide bond. Among these chains, a portion to which IgE binds is the α chain (FcεRIα). FcεRIα has a size of about 60 kDa, and is composed of a hydrophobic domain existing inside the cell membrane and a hydrophilic domain existing outside the cell membrane. In particular, IgE binds to an extracellular domain of the α chain. The FcεRIα can be interchangeably used as the alpha subunit of the IgE Fc receptor.

(25) Specifically, the alpha subunit of the IgE Fc receptor may have the amino acid sequence set forth in NP_001992.1. In addition, the extracellular domain (FcεRIα-ECD) of the alpha subunit of the IgE Fc receptor may have the amino acid sequence of SEQ ID NO: 1. The FcεRIα-ECD may be a fragment or variant of the FcεRIα-ECD, as long as the fragment or variant is capable of binding to IgE. In addition, the FcεRIα-ECD of SEQ ID NO: 1 may be encoded by a polynucleotide having the sequence of SEQ ID NO: 5.

(26) The variant may be prepared through a method of substituting, deleting, or adding one or more amino acids in the wild-type FcεRIα-ECD, as long as the method does not alter a function of the FcεRIα-ECD. Such variant may be 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more identical to the amino acid sequence of SEQ ID NO: 1.

(27) Here, the Fc region may be the wild-type Fc region or a modified Fc region. In addition, as used herein, the term “modified Fc region” means a region in which a part of an Fc portion of an antibody has been modified. Here, the Fc region refers to a protein which contains heavy chain constant region 2 (CH2) and heavy chain constant region 3 (CH3) of an immunoglobulin, and does not contain variable regions of heavy and light chains and light chain constant region 1 (CH1) of an immunoglobulin. In particular, the modified Fc region may be obtained by substituting some amino acids in the Fc region or by combining different types of Fc regions. Specifically, the modified Fc region may have the amino acid sequence of SEQ ID NO: 2. In addition, the modified Fc region of SEQ ID NO: 2 may be encoded by a polynucleotide having the sequence of SEQ ID NO: 6.

(28) In addition, the modified Fc region may be sugar chains in a native form, or increased sugar chains relative to a native form. Immunoglobulin Fc sugar chains may be modified by conventional methods such as chemical methods, enzymatic methods, and genetic engineering methods using microorganisms. In addition, the modified Fc region may be a region that lacks antibody dependent cellular cytotoxicity (ADCC) and complement dependent cytotoxicity (CDC) functions due to having no binding site for Fcγ receptors (FcγR) or complement component 1q (C1q).

(29) In an embodiment, the modified Fc region and the FcεRIα-ECD may be linked via a hinge of an IgD. The hinge may be a hinge region derived from immunoglobulin IgD or a variant thereof. The hinge region of the native IgD is composed of 64 amino acids. The hinge region derived from the immunoglobulin IgD or a variant thereof may be composed of 20 to 60 consecutive amino acids, 25 to 50 consecutive amino acids, or 30 to 40 amino acids. Here, the hinge variant may be obtained by modifying some in the amino acid sequence of the hinge region of IgD in order to minimize generation of truncated forms during a protein production process.

(30) In an embodiment, the hinge region derived from the immunoglobulin IgD or a variant thereof may be composed of 30 or 49 amino acids. In addition, the hinge region derived from the immunoglobulin IgD or a

variant thereof may contain at least one cysteine.

(31) In an embodiment, the hinge region derived from the immunoglobulin IgD or a variant thereof may contain the following sequence:

(32) Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1 Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro (SEQ ID NO: 17), where Xaa1 may be Lys or Gly, and Xaa2 may be Glu, Gly, or Ser. Specifically, the hinge region derived from the immunoglobulin IgD or a variant thereof may have the amino acid sequence of SEQ ID NO: 3 or SEQ ID NO: 19, thereby minimizing generation of truncated forms during a protein production process.

(33) In another embodiment, the hinge region derived from the immunoglobulin IgD or a variant thereof may contain the following sequence:

(34) Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro (SEQ ID NO: 18), where Xaa3 may be Lys or Gly, and Xaa4 may be Glu, Gly, or Ser. Specifically, the hinge region derived from the immunoglobulin IgD or a variant thereof may have the amino acid sequence of SEQ ID NO: 4, thereby minimizing generation of truncated forms during a protein production process.

(35) In particular, in the hinge region derived from the immunoglobulin IgD or a variant thereof having the sequence of SEQ ID NO: 4, at least one of Thr's may be glycosylated. Specifically, among the amino acids of SEQ ID NO: 18, the 13.sup.th, 14.sup.th, 18.sup.th, or 19.sup.th Thr's may be glycosylated. Preferably, all of the four Thr's may be glycosylated. Here, the glycosylation may be O-glycosylation.

(36) The sugar chain attached to the glycoprotein pharmaceuticals is one of the main factors that determine the quality to play an important role in therapeutic efficacy, persistence in the body, targeting and immune response and the like. It was confirmed that a polypeptide dimer in which the end of sugar chain is not terminated with sialic acid was rapidly removed from the body.

(37) The term "sialic acid" used in the present invention may include N-acetylneuraminic acid (Neu5Ac) of Formula 1 below and N-glycolylneuraminic acid (Neu5Gc) of Formula 2 below.

(38) ##STR00001##

(39) In this case, in one embodiment, a polypeptide dimer having a high content of sialic acid may have a high content of N-acetylneuraminic acid.

(40) In addition, a content of sialic acid of a polypeptide dimer may be increased through a purification method. In addition, the content of sialic acid of the polypeptide dimer may be increased by producing a polypeptide dimer in a cell into which a sialic acid transferase gene is introduced.

(41) The polypeptide dimer having a high content of sialic acid may be characterized in that the molar ratio of sialic acid/polypeptide dimer is 8 or more. For example, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30. Preferably, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25.

(42) In an embodiment, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, or at least 21. In addition, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 8 to 30 or 12 to 25. In addition, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 10 to 25, 11 to 24 or 12 to 23. Also, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 13 to 22, 14 to 22, 15 to 22, 16 to 22, 17 to 22, 18 to 22, or 19 to 22. Here, the sialic acid may be N-acetylneuraminic acid.

(43) In addition, the sialic acid can bind to 8 positions on a monomer comprising an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor and an Fc region. In this case, the sialic acid can bind to at least 4 or more of the above positions. Specifically, the sialic acid can bind to 4, 5, 6, 7 or 8 positions on the monomer.

(44) As described above, the polypeptide dimer having a high content of sialic acid provided by the present invention may be in a form in which two monomers are bound to each other and each monomer is obtained by binding between an extracellular domain of an alpha subunit of an IgE Fc receptor and a modified Fc region. The polypeptide dimer having a high content of sialic acid may be in a form in which the same two monomers are bound to each other by cysteine located at a hinge site. In addition, the polypeptide dimer having a high content of sialic acid may be in a form in which two different monomers are bound to each other.

(45) In this case, each polypeptide monomer may comprise sialic acid at the same position, but may comprise sialic acid at a different position. In addition, if the molar ratio of sialic acid/polypeptide dimer is 8 or more, each monomer constituting the dimer may have a different molar ratio of sialic acid/polypeptide.

(46) In addition, the polypeptide dimer may contain two monomers different from each other. Specifically, it may be in a form in which one monomer may contain the extracellular domain of the alpha subunit of the IgE Fc receptor, and the other monomer contains a fragment of the extracellular domain of the alpha subunit of the IgE Fc receptor. Here, an embodiment of the monomer may be composed of the amino acid sequence represented by SEQ ID NO: 20, SEQ ID NO: 21, or SEQ ID NO: 22.

(47) In addition, the polypeptide dimer having a high content of sialic acid provided by the present invention exhibits a binding affinity to IgE which is 10- to 100-fold, 20- to 90-fold, 20- to 70-fold, 30- to 70-fold, or 40- to 70-fold higher than omalizumab, an anti-IgE antibody, and may preferably exhibit a binding affinity to IgE which is about 70-fold higher than omalizumab.

(48) Another aspect of the present invention provides a pharmaceutical composition for preventing or treating allergic diseases, comprising the polypeptide dimer having a high content of sialic acid.

(49) The polypeptide dimer is the same as described above. In particular, the pharmaceutical composition may be characterized by being for subcutaneous injection. It was confirmed that a polypeptide dimer having a high content of sialic acid is effective in treating and/or preventing allergic diseases compared to a polypeptide dimer having a low content of sialic acid upon subcutaneous injection.

(50) As used herein, the term "allergic disease" means a pathological symptom caused by an allergic reaction mediated by mast cell activation such as mast cell degranulation. For example, such allergic diseases may be the one selected from the group consisting of food allergy, atopic dermatitis, asthma, allergic rhinitis, allergic conjunctivitis, allergic dermatitis, chronic idiopathic urticaria and allergic contact dermatitis, but are not limited thereto. In particular, the allergic diseases may be IgE-mediated.

(51) In the pharmaceutical composition, the polypeptide dimer may be contained in any amount (effective amount) depending on use, formulation, blending purpose, and the like, as long as the polypeptide dimer can exhibit anti-allergic activity. A typical effective amount of the polypeptide dimer will be determined within a range of 0.001% by weight to 20.0% by weight based on a total weight of the composition. Here, the effective amount refers to an amount which is capable of inducing an anti-allergic effect. Such an effective amount can be determined by those ordinarily skilled in the art.

(52) The pharmaceutical composition may further contain a pharmaceutically acceptable carrier. Specifically, the pharmaceutical composition contains a pharmaceutically acceptable carrier, and may be made into an oral or parenteral formulation depending on a route of administration by a conventional method known in the art.

(53) In a case where the pharmaceutical composition of the present invention is made into an oral formulation, the pharmaceutical composition may be made into formulations such as powders, granules, tablets, pills, sugar coating tablets, capsules, liquids, gels, syrups, suspensions, and wafers, together with suitable carriers, in accordance with methods known in the art. Here, examples of suitable pharmaceutically acceptable carriers can include sugars such as lactose, glucose, sucrose, dextrose, sorbitol, mannitol, and xylitol, starches such as corn starch, potato starch, and wheat starch, celluloses such as cellulose, methylcellulose, ethylcellulose, sodium carboxymethyl cellulose, and hydroxypropylmethyl cellulose, polyvinyl pyrrolidone, water, methylhydroxybenzoate, propylhydroxybenzoate, magnesium stearate, mineral oil, malt, gelatin, talc, polyol, vegetable oil, and the like. In a case of being made into preparations, the preparations can be carried out, as necessary, by including diluents and/or excipients such as a filler, an extender, a binder, a wetting agent, a disintegrant, and a surfactant.

(54) In a case where the pharmaceutical composition of the present invention is made into a parenteral formulation, the pharmaceutical composition may be made into preparations in the form of injections, transdermal drugs, nasal inhalers, and suppositories, together with suitable carriers, in accordance with methods known in the art. In a case of being prepared into injections, sterilized water, ethanol, polyol such as glycerol and propylene glycol, or a mixture thereof may be used as a suitable carrier. For the carrier, isotonic solutions such as Ringer's solution, phosphate buffered saline (PBS) containing triethanolamine, sterile water for injection, and 5% dextrose, and the like may be preferably used.

(55) Preparation of the pharmaceutical composition is known in the art and specifically, reference can be made to Remington's Pharmaceutical Sciences (19th ed., 1995) and the like.

(56) A preferable daily dosage of the pharmaceutical composition of the present invention is ranged from 0.01 µg/kg to 10 g/kg, and preferably from 0.01 mg/kg to 1 g/kg, depending on the patient's condition, body weight, sex, age, disease severity, or route of administration. The pharmaceutical composition may be

administered once a day or several times a day when the dosage is high.

(57) The subject to which the composition of the present invention can be applied (prescribed) is a mammal and may be, for example, a human, a dog and a cat, with a human being particularly preferred but not limited thereto. The pharmaceutical composition may further comprise any compound or natural extract, on which safety has already been verified and which is known to have anti-allergic activity, for the purpose of raising and reinforcing the anti-allergic activity.

(58) The present inventors studied to develop a safe and effective therapeutic agent for allergic diseases. As a result, it was confirmed that a polypeptide dimer having a high content of sialic acid has excellent binding affinity with IgE compared to a polypeptide dimer having a low content of sialic acid (Table 3). In addition, it was confirmed that when a polypeptide dimer having a high content of sialic acid is administered subcutaneously to a subject suffering from an allergic disease, the concentration of IgE and MCPT-1 in the blood is effectively reduced compared to a polypeptide dimer having a low content of sialic acid, and the effect persists even after a predetermined time has passed (FIGS. 13 and 14).

(59) Accordingly, another aspect of the present invention provides a transdermal patch comprising the pharmaceutical composition. In addition, another aspect of the present invention provides a topical patch comprising the pharmaceutical composition.

(60) Another aspect of the present invention provides a food composition for ameliorating or alleviating allergic symptom, comprising the polypeptide dimer having a high content of sialic acid.

(61) The polypeptide dimer having a high content of sialic acid is the same as described above. In addition, the polypeptide dimer having a high content of sialic acid may be bound to appropriate delivery means for efficient delivery into the intestine.

(62) In addition, the food composition may be prepared in any form and may be, for example, prepared in the form of beverages such as tea, juice, carbonated beverage, and ionic beverage, processed dairy products such as milk and yogurt, health functional food preparations such as tablets, capsules, pills, granules, liquids, powders, flakes, pastes, syrups, gels, jellies, and bars, or the like.

(63) The food composition may fall within any product category in legal or functional classification as long as the food composition complies with the enforcement regulations at the time of being manufactured and distributed. For example, the food composition may be a health functional food according to the Health Functional Foods Act, or may fall within confectioneries, beans, teas, beverages, special-purpose foods, or the like according to each food type in the Food Code of Food Sanitation Act (standards and specifications for food, notified by Food and Drug Administration). With regard to other food additives that may be contained in the food composition of the present invention, reference can be made to the Food Code or the Food Additive Code according to the Food Sanitation Act.

(64) Another aspect of the present invention provides a method for treating or preventing allergic diseases, comprising a step of administering to a subject the polypeptide dimer having a high content of sialic acid. The polypeptide dimer having a high content of sialic acid is the same as described above.

(65) The subject may be a mammal, preferably, a human, a dog and a cat. Here, administration may be achieved orally or parenterally. Here, parenteral administration may be performed by methods such as subcutaneous administration, intravenous administration, mucosal administration, and muscular administration.

(66) The allergic diseases may be the one selected from the group consisting of food allergy, atopic dermatitis, asthma, allergic rhinitis, allergic conjunctivitis, allergic dermatitis, chronic idiopathic urticaria and allergic contact dermatitis.

(67) Another aspect of the present invention provides a use of the polypeptide dimer having a high content of sialic acid for preventing or treating allergic diseases.

(68) Hereinafter, the present invention will be described in more detail with reference to the following examples. However, the following examples are intended to merely illustrate the present invention, and the scope of the present invention is not limited only thereto.

Example 1. Preparation of Polypeptide Containing Fc ϵ RI α -ECD and Fc Region

(69) A C-terminal modified polypeptide of the extracellular domain (Fc ϵ RI α -ECD) of the alpha subunit of the IgE Fc receptor was prepared according to the method disclosed in U.S. Pat. No. 7,867,491.

(70) First, in order to express a protein (Fc ϵ RI α ECD-Fc1), a protein (Fc ϵ RI α ECD-Fc2), and a protein (Fc ϵ RI α ECD-Fc3), in which the extracellular domain of the α -chain of Fc ϵ RI having the amino acid sequence of SEQ ID NO: 1 and the modified immunoglobulin Fc of SEQ ID NO: 2 are linked via a hinge of SEQ ID NO: 19, a hinge of SEQ ID NO: 3, and a hinge of SEQ ID NO: 4, respectively, cassettes obtained by linking

the gene encoding each protein were cloned into the pAD15 vectors (Genexin, Inc.) to construct FcεRIαECD-Fc protein expression vectors. Then, each of the expression vectors was transduced into CHO DG44 cells (from Dr. Chasm, Columbia University, USA).

(71) Here, at the time of being transduced into the cell line, an expression vector obtained by cloning an α-2,6-sialic acid transferase gene into the pCI Hygro vector (Invitrogen) was simultaneously transduced to separately prepare cell lines which are capable of expressing FcεRIαECD-Fc2ST and FcεRIαECD-Fc3ST proteins to which sialic acid is added.

(72) As a primary screening procedure, HT selection was carried out using 5-hydroxytryptamine (HT)-free 10% dFBS medium (Gibco, USA, 30067-334), MEMα medium (Gibco, 12561, USA, Cat No. 12561-049), and HT+ medium (Gibco, USA, 11067-030). Then, methotrexate (MTX) amplification was performed using HT-selected clones to amplify productivity using the dihydrofolate reductase (DHFR)-system.

(73) After completion of the MTX amplification, subculture was carried out about 1 to 5 times for cell stabilization for the purpose of evaluation of productivity. Thereafter, unit productivity evaluation of the MTX-amplified cells was performed. The results are shown in Table 1 below.

(74) TABLE-US-00001

TABLE 1	Productivity	MTX Batch	concen-	3-day culture	culture	Version	Media
tration	ug/mL	ug/10.sup.6 cells	(mg/ml)	FcεRIαECD-Fc2	Ex-	500 nM	37.23 20.9 225 FcεRIαECD-
cellDHFR	100 nM	45.4 25.1 338.2	Fc2 + a2,6-ST	FcεRIαECD-Fc3	2 uM	27.0 16.9 180.4	FcεRIαECD-
uM	17.5 10.2 101.7	Fc3 + a2,6-ST					

(75) As shown in Table 1, the FcεRIαECD-Fc3 cell line exhibited productivity of 16.9 μg/10.sup.6 cells after the methotrexate amplification at 2 uM. On the other hand, the FcεRIαECD-Fc3 cell line (FcεRIαECD-Fc3ST) co-transduced with 2,6-sialic acid transferase exhibited productivity of 10.2 μg/10.sup.6 cells after the methotrexate amplification at 1 uM. In addition, the FcεRIαECD-Fc2 cell line exhibited productivity of 20.9 μg/10.sup.6 cells under the methotrexate amplification condition at 0.5 uM. In addition, the FcεRIαECD-Fc2 cell line (FcεRIαECD-Fc2ST) co-transduced with 2,6-sialic acid transferase exhibited productivity of 25.1 μg/10.sup.6 cells after the methotrexate amplification at 0.1 uM. That is, it was identified that the FcεRIαECD-Fc2 cell line co-transduced with 2,6-sialic acid transferase, which had been selected under the methotrexate amplification condition at 0.1 uM, exhibits the most excellent productivity.

(76) The polypeptide (FcεRIαECD-Fc) produced from the FcεRIαECD-Fc2 cell line is referred to as "FcεRIαECD-Fc2," and the polypeptide produced from the FcεRIαECD-Fc2+a2,6-ST cell line is referred to as "FcεRIαECD-Fc2ST". In addition, the polypeptide produced from the FcεRIαECD-Fc3 cell line is referred to as "FcεRIαECD-Fc3," and the polypeptide produced from the FcεRIαECD-Fc3+a2,6-ST cell line is referred to as "FcεRIαECD-Fc3ST".

Example 2. Confirmation of Purification and Purity of Polypeptide (FcεRIα ECD-Fc)

(77) Among the cell lines selected in Example 1 above, i) FcεRIαECD-Fc3 cell line, ii) FcεRIαECD-Fc3+a2,6-ST cell line, and iii) FcεRIαECD-Fc2+a2,6-ST cell line were cultured at a 60 ml scale by a batch culture method. The resulting cultures were purified using a Protein-A affinity column to obtain FcεRIαECD-Fc, and then the purified FcεRIα ECD-Fc were subjected to SE-HPLC(Size-Exclusion High performance liquid chromatography) and SDS-PAGE to identify purity of the polypeptide.

(78) Specifically, SDS-PAGE was performed under a non-reducing condition. In the non-reducing condition, each purified polypeptide was mixed with a non-reducing sample buffer, and then electrophoresis was performed for 30 minutes in a Mini-Protean TGX™ gels (Bio-Rad) in a TGS (Tris Glycine SDS) buffer under a 200V condition. After electrophoresis, a protein was stained with a Coomassie Brilliant Blue solution. The results are shown in Table 2 and FIG. 1 below:

(79) TABLE-US-00002

TABLE 2	Purity Purifica-	(SE- Lane # Sample	tion HPLC)	Sample	condition
Protein standard	One-step	— — —	1 FcεRIαECD-Fc3 (Protein-A	94.5%	— Non-
affinity	93.7%	reduc-	3 FcεRIαECD-Fc2ST column)	93.2%	ing 4 FcεRIαECD-Fc3 Purifica-
5 FcεRIαECD-Fc3ST	93.7%	Thawing	6 FcεRIαECD-Fc2ST	93.2%	test

(80) As shown in Table 2 above, it was confirmed that the purity of each polypeptide purified by the SE-HPLC method was 93% or more. It was confirmed that impurities such as a truncated form did not appear in a non-reducing condition, and in particular, the purity was 93% or more and there were no impurities even after the thawing/freezing process.

Experimental Example 1. Confirmation of Dimer Formation of Polypeptide (FcεRIα ECD-Fc)

(81) Among the cell lines selected in Example 1 above, i) FcεRIαECD-Fc3 cell line, ii) FcεRIαECD-Fc3+a2,6-ST cell line, and iii) FcεRIαECD-Fc2+a2,6-ST cell line were cultured at a 60 ml scale by a batch culture method. Thereafter, SE-HPLC and SDS-PAGE were performed on the purified product obtained from

the purification of polypeptide using the culture supernatant and Protein-A affinity column. In this case, they were performed on the culture supernatant and the purified product under a non-reducing condition and a reducing condition, respectively.

(82) The non-reducing condition was performed in the same manner as in Example 2. On the other hand, in a reducing condition, each purified polypeptide was mixed with a reducing sample buffer comprising 2-mercaptoethanol, and then denatured at a temperature of 100° C. for 5 minutes. Thereafter, electrophoresis was performed for 30 minutes under a 200V condition using a TGS buffer in Mini-Protean TGX™ gels (Bio-Rad). After electrophoresis, a protein was stained with a Coomassie Brilliant Blue solution.

(83) As a result, a polypeptide having a size of about 150 kDa was detected under a non-reducing condition, and a polypeptide having a size of about 75 kDa was detected under a reducing condition. Through this, it was confirmed that the polypeptide forms a dimer (FIG. 2). In particular, it was confirmed that the purity of a polypeptide dimer is high even in the culture supernatant corresponding to the input, and no polypeptide was detected in the sample (FT; Flow Through) obtained by passing the culture supernatant through an affinity column, but a polypeptide was detected in the eluted sample. In addition, no significant difference was observed even when the sample (Elute N) that was immediately neutralized with 1 M Tris buffer (pH 9.0) due to the low pH of the eluted sample was compared with the sample (Elute) that was not neutralized (FIG. 2).

(84) In addition, it was identified that not only polypeptides with very high purity (98% or higher) are purified but also polypeptides are expressed with very high purity even in the culture supernatant. This indicates that process development steps can be simplified in developing the polypeptides, which had been expressed in the cell lines in question, into a medical product, and as a result, it is highly likely for the development cost of the medical product to be remarkably decreased.

Experimental Example 2. Confirmation of Binding Affinity of Polypeptide Dimer to IgE

(85) A binding affinity to IgE was comparatively measured for the purified products obtained by purifying the polypeptides which had been produced in Example 1, i) FcεRIαECD-Fc2, ii) FcεRIαECD-Fc2ST, iii) FcεRIαECD-Fc3, and iv) FcεRIαECD-Fc3ST, through the method of Example 2 above, and the commercially available anti-IgE antibody, omalizumab (trade name: Xolair).

(86) Specifically, the binding affinity to IgE was measured by coating IgE on the channel of the Protein GLC sensor chip (Bio-Rad Laboratories, Inc., Cat #176-5011), and causing omalizumab or each FcεRIαECD-Fc protein at various concentrations to flow at a rate of 30 μL per minute. Here, the experiments were conducted by identifying zero base using 25 mM NaOH as a regeneration buffer, and then repeating the above steps. Thereafter, a binding curve was identified using a protein binding analyzer (ProteOn XPR36, Bio-Rad Laboratories, Inc., USA). The results are shown in Table 3.

(87) TABLE-US-00003 TABLE 3 Samples items FcεRIα ECD-Fc Omalizumab Drug type Fc fusion protein Anti-IgE Ab Remarks Binding k_a Fc2 $2.14 \times 10^{sup.5}$ $4.05 \times 10^{sup.5}$ 1.9-fold weaker than omalizumab affinity (Association Fc2ST $2.64 \times 10^{sup.5}$ 1.5-fold weaker than omalizumab rate) Fc3 $1.98 \times 10^{sup.5}$ 2.0-fold weaker than omalizumab Fc3ST $2.40 \times 10^{sup.5}$ 1.7-fold weaker than omalizumab k_d Fc2 $8.29 \times 10^{sup.-5}$ $6.02 \times 10^{sup.-3}$ 73-fold better than omalizumab (Dissociation Fc2ST $5.69 \times 10^{sup.-5}$ 106-fold better than omalizumab rate) Fc3 $1.33 \times 10^{sup.-4}$ 45-fold better than omalizumab Fc3ST $1.49 \times 10^{sup.-4}$ 40-fold better than omalizumab KD (k_d/k_a) Fc2 $3.88 \times 10^{sup.-10}$ $1.49 \times 10^{sup.-8}$ 38-fold better than omalizumab Fc2ST $2.16 \times 10^{sup.-10}$ 69-fold better than omalizumab Fc3 $6.72 \times 10^{sup.-10}$ 22-fold better than omalizumab Fc3ST $6.21 \times 10^{sup.-10}$ 24-fold better than omalizumab

(88) As shown in Table 3, the association rate (k_a) value of the polypeptide dimer according to an embodiment of the present invention and IgE was measured to be 1.5- to 2.0-fold lower than that of omalizumab. That is, it was found that a binding affinity thereof to substances other than IgE is 1.5- to 2.0-fold lower than that of omalizumab. In addition, the dissociation rate (k_d) value of the polypeptide dimer according to an embodiment of the present invention was measured to be 40- to 106-fold higher than that of omalizumab. As a result, it was confirmed that the polypeptide dimer according to an embodiment of the present invention has an equilibrium dissociation constant ($KD < k_d/k_a >$) value which is 22- to 69-fold higher than omalizumab. From this, it was identified that the polypeptide dimer according to an embodiment of the present invention has a remarkably increased binding affinity to IgE as compared with omalizumab. In particular, it was identified that the polypeptide dimer to which sialic acid is added (FcεRIαECD-Fc2ST) exhibits the highest IgE-binding affinity which is 69-fold higher than omalizumab.

(89) In addition, as shown in FIGS. 3 and 4, it was confirmed that omalizumab loses its binding to IgE when a certain period of time has passed after the binding, whereas the polypeptide dimer according to an embodiment of the present invention (FcεRIαECD-Fc2ST, IgE.sub.TRAP) is not separated from IgE once it

binds to IgE. That is, it is confirmed that the polypeptide dimer according to an embodiment of the present invention is not easily separated from IgE, and has a much better ability to maintain its bound state than omalizumab.

Experimental Example 3. Confirmation of Binding Affinity of Polypeptide Dimer to IgG Receptors

(90) The binding affinity of a polypeptide dimer according to an embodiment of the present invention (Fc ϵ RI α ECD-Fc2ST, IgE.sub.TRAP) or omalizumab (Xolair) to Fc gamma receptors of IgG was confirmed using Octet RED384 system (Pall ForteBio, CA, USA).

(91) Specifically, Fc γ RI, Fc γ RIIA, Fc γ RIIB, Fc γ RIIIA, and Fc γ RIIIB recombinant proteins (R & D Systems Inc., 5 μ g/ml) which are Fc gamma receptors were immobilized in 300 mM acetate buffer (pH 5) on the activated AR2G biosensor. As a running buffer, PBS containing 0.1% Tween-20 and 1% bovine serum was used. All measurements were carried out at 30° C. with a sample plate shaker at a rate of 1,000 rpm. The results are shown in FIGS. 5a to 5e, and binding affinity of omalizumab and the polypeptide dimer to the IgG Fc gamma receptors are quantified and shown in FIG. 6.

(92) As a result, it was confirmed that omalizumab showed a high binding affinity to Fc gamma receptors of IgG, whereas the polypeptide dimer had a significantly low binding affinity to Fc gamma receptors of IgG. Through this, it was confirmed that the polypeptide dimer does not bind with Fc gamma receptors of IgG.

Experimental Example 4. Confirmation of Activity of Polypeptide Dimer Through Beta-Hexosaminidase Assay in Mouse Bone Marrow-Derived Mast Cells

(93) Beta-hexosaminidase assay was performed for in vitro activity analysis of the polypeptide dimer according to an embodiment of the present invention.

(94) Specifically, the polypeptide dimer (Fc ϵ RI α ECD-Fc2) according to an embodiment of the present invention was mixed, at each concentration, with IgE (1 μ g/mL), and incubated at 20° C. for 30 minutes to prepare samples. Mouse bone marrow-derived mast cells in culture for mast cell activation were washed with Hank's balanced salt solution (HBSS) buffer to remove the medium, the number of cells was measured, and then 5 $\times$ 10⁵ cells were refloated into 40 μ L of HBSS buffer.

(95) 50 μ L of the sample thus prepared was added to activated mast cells. Then, the resultant was incubated in a 5% CO₂ incubator at 37° C. for 30 minutes. Subsequently, after the addition of each 10 μ L of DNP (2,4-dinitrophenol, 100 ng/mL), which is a foreign antigen, incubation was performed again at 37° C. for 30 minutes in 5% CO₂, and then 30 μ L of the supernatant was separated.

(96) 30 μ L of the separated supernatant and 30 μ L of the substrate (4-nitrophenyl N-acetyl- β -D-glucosaminide, 5.84 mM) were mixed well, and then incubated at 37° C. for 20 minutes in 5% CO₂. Then, 140 μ L of 0.1 M sodium carbonate buffer (pH 10) as a stop solution was added to terminate the reaction. Thereafter, absorbance at 405 nm was measured to identify a secretion amount of β -hexosaminidase secreted by the foreign antigen in the activated mast cells. The results are shown in FIG. 7.

(97) As shown in FIG. 7, it was confirmed that the activity of IgE is considerably inhibited from the case in which the concentration of IgE and the polypeptide dimer of the present invention is 1:1. That is, it was confirmed that the polypeptide dimer according to an embodiment of the present invention reacts even when the concentration thereof is the same as that of IgE.

Experimental Example 5. Comparison of Activity of Polypeptide Dimer and Human Anti-IgE Antibody Using β -Hexosaminidase Assay in Human Fc ϵ RI-Expressing Mouse Bone Marrow-Derived Mast Cells

(98) In order to compare the activity of the polypeptide dimer according to an embodiment of the present invention (Fc ϵ RI α ECD-Fc2ST, IgE.sub.TRAP) and Xolair (omalizumab), which is a human anti-IgE antibody, a beta-hexosaminidase analysis was performed.

(99) The respective polypeptide dimers and Xolair were prepared at each concentration, and then mixed with human IgE (1 μ g/mL). Then, incubation was performed at room temperature for 30 minutes to prepare samples. In addition, mast cells derived from and differentiated from mouse bone marrow, in which a human Fc ϵ RI gene had been introduced and the mouse Fc ϵ RI gene had been removed, were prepared. The prepared mast cells were washed with HBSS buffer, and then 5 $\times$ 10⁵ cells were refloated into 60 μ L of HBSS buffer.

(100) Then, 20 μ L of the sample thus prepared was added to the mast cells, and then incubated in a 5% CO₂ incubator at 37° C. for 30 minutes. Subsequently, after 20 μ L of human anti-IgE antibody (BioLegend, Cat No. 325502, 0.5 μ g/mL) was added, the resultant was incubated again in 5% CO₂ incubator at 37° C. for 30 minutes. Then, after centrifugation at 1,500 rpm at 4° C., 30 μ L of the supernatant was separated. 30 μ L of the separated supernatant and 30 μ L of the substrate (4-nitrophenyl N-acetyl- β -glucosaminide, 5.84 mM) were mixed well, and then incubated in a 5% CO₂ incubator at 37° C. for 25

minutes. Then, 140 μ L of 0.1 M sodium carbonate buffer (pH 10) was added to terminate the reaction. Thereafter, absorbance at 405 nm was measured to compare relative amounts of secreted β -hexosaminidase, and an inhibitory effect against mast cells was identified depending on each concentration of the sample. The results are shown in FIG. 8.

(101) As shown in FIG. 8, IC_{sub.50} of the polypeptide dimer was measured to be approximately 11.16 ng/mL, and IC_{sub.50} of Xolair was measured to be approximately 649.8 ng/mL. Therefore, it was identified that the polypeptide dimer has a 58-fold higher inhibitory ability on mast cell activity than Xolair.

Experimental Example 6. Confirmation of Activity of Polypeptide Dimer Through In Vivo Assay in Food Allergy Model

(102) 50 μ g of ovalbumin (OVA) and 1 mg of alum were intraperitoneally administered to Balb/c mice (Orientbio Inc.) two times at a 14-day interval to induce sensitization. Thereafter, 50 mg of OVA was orally administered five times in total on days 28, 30, 32, 34, and 36, to induce food allergy in intestines.

(103) After the OVA was orally administered two times, that is, on day 31, the mice were divided into three groups, each containing 7 mice. The three divided groups were as follows: Group 1 receiving the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) at a high concentration (200 μ g), Group 2 receiving the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) at a low concentration (20 μ g), and Group 3 receiving nothing.

(104) While orally administering the OVA, it was observed whether diarrhea occurs due to food allergy induction. In addition, the mice were sacrificed on day 37, and the number of mast cells in the small intestine, the IgE concentration in blood, and the concentration of the mast cell degranulation enzyme (mast cell protease-1 (MCPT-1)) in blood were analyzed for the mice belonging to each group.

(105) As a result, as shown in FIG. 9, diarrhea occurred in mice of Group 3 after the second oral administration of OVA. On the other hand, diarrhea occurred in mice of Group 1 and Group 2 after the third oral administration of OVA. In particular, the frequency of diarrhea in mice of Group 1 was lower than that in mice of Group 2. Through this, it was confirmed that the effect against food allergy increased in proportion to the concentration of a polypeptide dimer.

Experimental Example 7. Analysis of Content of Sialic Acid of Polypeptide Dimer

(106) Based on the aspects that the production rate of the polypeptide produced from the Fc ϵ RI α ECD-Fc2+a2,6-ST cell line prepared in Example 1 above is the highest, and the anti-allergic effect of the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) in Experimental Examples 4 to 6 is excellent, the sialic acid content of the polypeptide dimer was analyzed to investigate the anti-allergic efficacy of the polypeptide dimer depending on the sialic acid content.

(107) Specifically, in order to measure the sialic acid content contained in the glycan structure of the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) produced from the Fc ϵ RI α ECD-Fc2+a2,6-ST cell line prepared in Example 1 above, sialic acid was first separated by treatment with sialidase, which is an enzyme associated with sialic acid catabolism. Thereafter, the separated sialic acid was isolated, detected and quantified using HPLC (waters, alliance e2659).

(108) The polypeptide dimer was divided into three samples according to the pH gradient, and isolated and purified. Three samples were placed in an AmiconUltra 10K (Millipore, UFC501096) filter, and centrifuged for 10 minutes under a condition of 13,000 rpm, 4° C. temperature, and repeated 5 times, and the concentrate was exchanged with deionized water and concentrated. The concentration of the sample was set to 10 mg/mL or more when measured at a wavelength of 280 nm.

(109) Thereafter, 0.5 mg of each sample was taken and placed in an EP tube, and 1 μ L of sialidase (Roche, 10 269 611 001) and 40 μ L of sodium phosphate buffer (pH 7.0) at a concentration of 10 mM were added, and then deionized water was added so that the final volume was 100 μ L. 2 μ L of each sample was taken to measure the concentration at a wavelength of 280 nm, and the confirmed concentration value was used as the final analysis concentration.

(110) The sample was reacted for 18 hours in a 37° C. incubator, and then placed in an AmiconUltra 10K filter, and centrifuged for 15 minutes under a condition of 13,000 rpm, 4° C., and the filtrate passing through a filter was used for analysis. HPLC analysis conditions are shown in Table 4 below.

(111) TABLE-US-00004 TABLE 4 Analysis Column RHM-monosaccharide H⁺ (8%) 300 $\times$ 7.8 mm (Rezex); analysis column RHM-monosaccharide H⁺ (8%) 50 $\times$ 7.8 mm (Rezex); guard column Standard Substance NGNA: 2~40 μ M NANA: 100~2000 μ M Flux 0.55 ml/min Column Temperature 50° C. Detection 206 nm Injection Volume 5 μ L Mobile Phase 5 mN sulfuric acid in water Gradient/Time Isocratic Elution/45 minutes

(112) The standard substance for calculating the content of sialic acid is a mixture of N-acetylneuraminic acid (hereinafter referred to as NANA) and N-glycolylneuraminic acid (hereinafter referred to as NGNA). The

linear regression equation for the standard curve was obtained to calculate the molar concentrations of NGNA and NANA in the analyte. The mixtures used are shown in Table 5 below.

(113) TABLE-US-00005 TABLE 5 Mixture 1 2 3 4 5 NANA (μM) 2,000 1,000 500 300 200 NGNA (μM) 40 20 10 6 4

(114) The NANA sialic acid content in the sample was calculated using the NANA content. The sialic acid content in the sample was expressed as a molar ratio of sialic acid/sample (mol/mol).

(115)
$$\text{Sialic acid content of Sample} = (\text{molar concentration of NANA in sample}) / (\text{Molar concentration of sample}) \quad [\text{Equation 1}]$$

(116) In addition, the content of NGNA in the sample was calculated using the NGNA content. The content of NGNA in the sample was expressed as a molar ratio of NGNA/analyte (mol/mol).

(117)
$$\text{NGNA Content of sample} = (\text{NGNA molar concentration in analyte}) / (\text{Molar concentration of analyte}) \quad [\text{Equation 2}]$$

(118) The results are shown in Table 6 below.

(119) TABLE-US-00006 TABLE 6 NGNA/Protein NANA/Protein Sample (mol/mol) (mol/mol) Fc ϵ RI α ECD-Fc2ST Sample 1 (SA.sup.low) 0.13 7.7 Sample 2 (SA.sup.medi) 0.17 12.0 Sample 3 (SA.sup.high) 0.27 19.1

(120) As shown in Table 6, it was divided into three samples according to the pH gradient, and separated and purified, and the sialic acid content in the samples was measured differently. In order to compare the anti-allergic efficacy according to the sialic acid content, a polypeptide dimer is referred to as “SA.sup.low”, “SA.sup.medi” and “SA.sup.high” in order of the sialic acid content.

Experimental Example 8. Confirmation of Change in Dimer Formation of Polypeptide According to Sialic Acid Content

(121) In order to confirm whether there was a change in dimer formation of the polypeptide according to the sialic acid content, SA.sup.low, SA.sup.medi and SA.sup.high separated in Example 9 were subjected to SDS-PAGE analysis under a non-reducing condition and a reducing condition in the same manner as in Experimental Example 1.

(122) As a result, in all SA.sup.low, SA.sup.medi and SA.sup.high, a polypeptide having a size of about 150 kDa was detected under a non-reducing condition, and a polypeptide having a size of about 75 kDa was detected under a reducing condition. Through this, it was confirmed that the polypeptide forms a dimer (FIG. 10).

Experimental Example 9. Confirmation of Isoelectric Point of Polypeptide Dimer According to Sialic Acid Content

(123) In order to confirm whether the isoelectric point of the polypeptide dimer changes according to the sialic acid content, SA.sup.low, SA.sup.medi and SA.sup.high Separated in Experimental Example 7 were mixed with the IEF sample solution, and then loaded into a pH 3-7 IEF gel (Invitrogen), and electrophoresis was sequentially performed for 1 hour under a 100V condition, for 1 hour under a 200V condition, and for 30 minutes under a 500V condition. After the electrophoresis was completed, the gel was reacted in a fixing solution comprising 12% trichloroacetic acid and 3.5% sulfosalicylic acid for 30 minutes, and then washed with distilled water. A protein was stained with a Coomassie Brilliant Blue solution.

(124) As a result, the isoelectric points of SA.sup.low, SA.sup.medi and SA.sup.high were slightly different according to the sialic acid content, and were found to be about 5.3, 4.9 and 4.7, respectively (FIG. 11).

Experimental Example 10. Confirmation of Anti-Allergic Activity of Polypeptide Dimer According to Sialic Acid Content in Food Allergy Animal Model: Measurement of Frequency of Diarrhea

(125) In order to confirm whether the activity of the polypeptide dimer changes according to the sialic acid content, sensitization was induced by intraperitoneal administration of 50 μg of OVA and 1 mg of alum twice at an interval of 14 days for Balb/c mice (Orient Bio). Thereafter, at Day 28, 30, 32, 34, 36, 38, 40 over a total of 7 times, food allergy was induced by oral administration of 50 mg of OVA. In this case, the oral administration of OVA was performed after fasting for 4 hours.

(126) The OVA was administered orally twice, and then at Day 40, the mice were divided into 3 groups of 7 mice. The mice were divided into Group 1, a group in which a polypeptide dimer having a high content of sialic acid (SA.sup.high) is subcutaneously administered at a high concentration (200 μg), Group 2, a group in which a polypeptide dimer having a low content of sialic acid (SA.sup.low) is subcutaneously administered at a high concentration (200 μg), and Group 3, a group in which PBS is subcutaneously administered. It was confirmed whether diarrhea occurred according to the induction of food allergy when the OVA was administered orally.

(127) As a result, diarrhea occurred in mice of Group 3 after the fourth oral administration of OVA. Diarrhea

occurred in mice of Group 2 after the fifth oral administration of OVA. In particular, the frequency of diarrhea increased rapidly in mice of Group 2 after the sixth oral administration of OVA. On the other hand, diarrhea occurred in mice of Group 1 after the sixth oral administration of OVA, but the frequency of diarrhea was maintained but did not increase in mice of Group 1 even after the seventh oral administration of OVA (FIG. 12). Through this, it was confirmed that the polypeptide dimer having a high content of sialic acid (SA.sup.high) had an excellent anti-allergic effect compared to the polypeptide dimer having a low content of sialic acid (SA.sup.low).

Experimental Example 11. Confirmation of Anti-Allergic Activity of Polypeptide Dimer According to Sialic Acid Content in Food Allergy Animal Model: Measurement of IgE Concentration in Blood

(128) The blood was collected by orbital sampling method from mice of each group in which the seventh oral administration of OVA of Experimental Example 10 was performed. After reacting for 30 minutes at ambient temperature, serum was separated by centrifugation for 15 minutes under a condition of 13,000 rpm at a temperature of 4° C. In order to measure free IgE concentration in the blood, mouse total IgE ELISA kit (BioLegend) was used, and in this case, ELISA was performed according to the manufacturer's protocol except that a polypeptide dimer (SA.sup.low and SA.sup.high) according to an embodiment of the present invention was used as a substance coated on a 96-well-plate instead of an anti-IgE antibody.

(129) Specifically, a 96-well-plate was coated with a polypeptide dimer (SA.sup.low and SA.sup.high) diluted with PBS, and reacted overnight at a temperature of 4° C. The next day, after washing with PBS containing 0.05% tween 20 (hereinafter, washing buffer), a blocking buffer (assay diluent) was added and reacted for 1 hour. Thereafter, it was washed with a washing buffer, and then the mouse IgE to be used as a standard solution and the serum samples of mice were diluted in a 1× assay diluent, and placed in a plate and reacted for 2 hours.

(130) After washing again with a washing buffer, biotin-labeled mouse anti-IgE antibody was added and reacted for 1 hour. After washing with a washing buffer, HRP (horse radish peroxidase)-labeled avidin (Avidin-HRP) was added and reacted for 30 minutes. After washing with a washing buffer, a substrate solution was added and reacted for 20 minutes while blocking light, and then a stop solution (1 M H.sub.2SO.sub.4) was added to stop the reaction. Thereafter, the absorbance value was measured at a wavelength of 450 nm with a microplate reader (Epoch Microplate Spectrophotometer), and the concentration was calculated.

(131) As a result, for the mice of Group 3 in which PBS was subcutaneously administered, the IgE concentration in the blood was calculated to be about 8,000 ng/mL. In addition, for the mice of Group 2 in which a polypeptide dimer having a low content of sialic acid (SA.sup.low) was subcutaneously administered, the IgE concentration in the blood was calculated to be about 7,000 ng/mL. On the other hand, for the mice of Group 1 in which a polypeptide dimer having a high content of sialic acid (SA.sup.high) was subcutaneously administered, the IgE concentration in the blood was calculated to be about 4,900 ng/mL, and there was a significant difference from the IgE concentration values in the blood between mice of Group 2 and Group 3 (FIG. 13).

(132) From these results, it was found that when the polypeptide dimer having a high content of sialic acid was administered subcutaneously, the IgE content in the blood was reduced. In addition, based on such reduction of IgE in the blood, it was predicted that the polypeptide dimer having a high content of sialic acid is effective in treating allergy.

Experimental Example 12. Confirmation of Anti-Allergic Activity of Polypeptide Dimer According to Sialic Acid Content in Food Allergy Animal Model: Measurement of MCPT-1 Concentration in Blood

(133) Serum was prepared by taking blood from mice of each group 2 days after the food allergy experiment was completed. In order to measure the MCPT-1 (Mast Cell Protease-1) concentration in the blood, an MCPT-1 ELISA kit (Invitrogen) was used to perform the procedure according to the manufacturer's protocol.

(134) Specifically, a 96-well-immunized plate was coated with a mouse anti-MCPT-1 antibody and reacted overnight at a temperature of 4° C. The next day, after washing with PBS containing 0.05% tween 20 (hereinafter, washing buffer), PBS containing 1% BSA (bovine serum albumin) (hereinafter, blocking buffer, assay diluent) was added and reacted for 1 hour. Thereafter, it was washed with a washing buffer, and then the MCPT-1 to be used as a standard solution and the serum samples of mice were diluted in a 1× assay diluent, and placed in a plate and reacted for 2 hours.

(135) After 2 hours, a biotin-labeled mouse anti-MCPT-1 antibody was added and reacted for 1 hour. After washing with a washing buffer, HRP (horse radish peroxidase)-labeled avidin (Avidin-HRP) was added and reacted for 30 minutes. After washing with a washing buffer, a substrate solution was added and reacted for 20

minutes while blocking light, and then a stop solution (1 M H.sub.2SO.sub.4) was added to stop the reaction. Thereafter, the absorbance value excluding that measured at a wavelength of 570 nm wavelength from that measured at a wavelength of 450 nm was determined with a microplate reader (Epoch Microplate Spectrophotometer) and the concentration was calculated.

(136) As a result, for the mice of Group 3 in which PBS was subcutaneously administered, the MCPT-1 concentration in the blood was calculated to be about 4,000 ng/mL. In addition, for the mice of Group 2 in which a polypeptide dimer having a low content of sialic acid (SA.sup.low) was subcutaneously administered, the MCPT-1 concentration in the blood was calculated to be about 4,200 ng/mL. On the other hand, for the mice of Group 1 in which a polypeptide dimer having a high content of sialic acid (SA.sup.high) was subcutaneously administered, the MCPT-1 concentration in the blood was calculated to be about 2,800 ng/mL, and there was a significant difference from the MCPT-1 concentration values in the blood between mice of Group 2 and Group 3 (FIG. 14).

Example 3. Preparation of Polypeptide Dimer Having High Content of Sialic Acid Through Purification

(137) In order to confirm the physical properties of a polypeptide dimer according to the sialic acid content, a polypeptide dimer containing a wide range of sialic acid was obtained. Specifically, a polypeptide dimer obtained from the FcεRIαECD-Fc2+a2,6-ST cell line was purified using affinity chromatography and anion exchange chromatography.

(138) First, the cell culture solution was subjected to affinity chromatography to remove the primary impurities from the culture solution. The affinity chromatography was performed using Amsphere™ A3 pre-packing column (bed height 5 cm, volume 5 mL) packed with Amsphere™ A3 resin and AKTA Avant 25 equipment, a liquid chromatography system. Here, it was performed using 100 mM Glycine, pH 3.3 as an eluted buffer.

(139) Thereafter, the anion exchange chromatography was performed using Hiscreen QFF (bed height 10 cm, volume 5 mL) packed with Q Sepharose Fast Flow resin and AKTA Avant 25 equipment, a liquid chromatography system. 20 mM phospho-citrate, pH 3.5 was used as an eluted buffer. In this case, the concentration of 20 mM phospho-citrate, pH 4.0, as a washing buffer, was adjusted to obtain dimers having various sialic acid contents. Analysis of the sialic acid content was performed by the analysis method used in Experimental Example 7. Through this, polypeptide dimers having a sialic acid content of 7.0 mol/mol, 10.3 mol/mol, 12.9 mol/mol, 14.9 mol/mol and 21.4 mol/mol were obtained.

Experimental Example 13. Pharmacokinetics Analysis of Polypeptide Dimer According to Sialic Acid Content in Mouse Model

(140) In order to perform pharmacokinetics analysis according to the sialic acid content of the polypeptide dimer, the polypeptide dimers having a sialic acid content of 7.0 mol/mol, 10.3 mol/mol, 12.9 mol/mol, 14.9 mol/mol and 21.4 mol/mol prepared in Example 3 was subcutaneously injected into mice at a concentration of 10 mg/kg, respectively. In order to analyze the pharmacokinetics of the polypeptide dimer, the blood was collected at Hour 0, Hour 3, Hour 10, Hour 24, Hour 72, Hour 168, and Hour 240 after administration of the substance and analyzed in the following manner.

(141) The anti-FcεRI antibody (Invitrogen) was diluted in 1×PBS (Welgene) at a concentration of 0.5 μg/mL, and added to an immunoplate (Thermo), and reacted overnight under a condition of 4° C. After the reaction was completed, the immunoplate was washed with 1×PBST, and I-Block solution (Invitrogen), a blocking solution, was added per well and reacted at ambient temperature for 1 hour. During blocking, standard and analytical samples were prepared. The polypeptide dimer was diluted in 1% BSA/PBS containing mouse blank serum at a concentration of 0.19 ng/mL to 200 ng/mL to prepare a standard substance, and the dilution with 1/20-1/300 using 1% BSA/PBS containing the blank serum was also performed to prepare an analytical sample.

(142) After the blocking was completed, the plate was washed with 1×PBST, and the prepared standard and analytical samples were added per well, and then reacted at ambient temperature for 1 hour. Thereafter, it was washed with 1×PBST, and an anti-human IgG4-HRP (Southern biotech) diluted to 1:10,000 was added, and then reacted at ambient temperature for 1 hour. After the reaction was completed, the plate was washed with 1×PBST, and TMB solution (Thermo) was added to the well and reacted. When the absorbance value was measured in the range of 0.8-1.0 at a wavelength of 650 nm using a microplate reader (Molecular devices), a stop solution (Sigma) was added per well to stop the reaction. After the reaction was terminated, the absorbance was measured in the plate at a wavelength of 450 nm within 5 minutes and the numerical values were analyzed. The resulting pharmacokinetic graph and pharmacokinetic parameters of the polypeptide dimer according to the sialic acid contents are shown in FIG. 15 and Table 7.

(143) As a result, as shown in Table 7, it was confirmed that the higher the sialic acid content of the polypeptide dimer administered to the mouse, the higher the concentration in the blood.

(144) TABLE-US-00007 TABLE 7 sialic acid content (mol/mol) T.sub.1/2 (hr) T.sub.max (hr) C.sub.max (μg/ml) AUC.sub.last (hr*μg/ml)

7.0	51.3 ± 5.9	5.3 ± 3.3	1.3 ± 0.5	47.8 ± 3.2	10.3	44.5 ± 6.7	5.3 ± 3.3
1.6 ± 0.6	57.1 ± 9.6	12.9	40.0 ± 9.5	17 ± 9.9	2.0 ± 0.4	119.7 ± 5.2	14.9
35.4 ± 3.5	7.7 ± 3.3	2.6 ± 0.3	182.7 ± 7.0	21.4	38.5 ± 5.0	10.0 ± 0.0	12.1 ± 0.9
890.0 ± 64.4							

Experimental Example 14. Confirmation of Efficacy According to Sialic Acid Content of Polypeptide Dimer in Passive Systemic Anaphylaxis Mouse Model

(145) In order to confirm the difference in an anti-allergic effect in vivo according to the sialic acid content of the polypeptide dimer, the method disclosed in the literature was applied to induce the passive systemic anaphylaxis (PSA) in mice in the manner as follows (Methods and Protocols, Methods in Molecular Biology, vol. 1032, DOI 10.1007/978-1-62703-496-8_10). It is known that mast cells are activated by induction of anaphylaxis and body temperature is reduced by increasing mast cells. While inducing PSA in mice using this, the polypeptide dimer having the respective sialic acid content was administered, and then the body temperature was measured to compare the anti-allergic effect.

(146) The body temperature was measured in transponder-inserted Balb/c mice (Orient Bio), and then 20 μg of anti-DNP IgE (Sigma) containing 0.9% NaCl was intraperitoneally administered, and then the polypeptide dimers having different sialic acid contents, respectively, were subcutaneously administered at 10 mg/kg. After 24 hours, the body temperature of the mice was measured before DNP-HAS (Sigma) containing 0.9% NaCl, a foreign antigen, was administered. Thereafter, 1 mg of the antigen was administered intravenously. After administration, the body temperature of the mice was measured at an interval of 10 minutes for 1 hour using a remote body temperature meter (Bio medic data systems). The experimental results are shown in FIG. 16.

(147) As a result, it was confirmed that the body temperature of the mice was reduced in groups in which a vehicle and a polypeptide dimer having a sialic acid content of 10.3 mol/mol were administered, and the body temperature of the mice was not reduced in groups in which the polypeptide dimer having a sialic acid content of 14.9 mol/mol or more was administered. Through this, it was confirmed that a polypeptide dimer having a high content of sialic acid has an excellent effect of reducing the induction of passive systemic anaphylaxis compared to a polypeptide dimer having a low content of sialic acid.

Claims

1. A polypeptide dimer comprising: a first monomer comprising an extracellular domain of an alpha subunit of an IgE Fc receptor (FcεRIα-ECD), a second monomer comprising an extracellular domain of an alpha subunit of an IgE Fc receptor (FcεRIα-ECD), wherein the FcεRIα-ECD of the first and second monomers comprises the amino acid sequence of SEQ ID NO: 1, wherein the first monomer and the second monomer each contain an Fc region, and wherein the Fc region and the FcεRIα-ECD, in the first and second monomers, are linked via a hinge, and wherein the polypeptide dimer comprises sialic acid and, a molar ratio of the sialic acid to the polypeptide dimer is 12 to 25.
2. The polypeptide dimer according to claim 1, wherein the Fc region of the first and second monomers comprises the amino acid sequence of SEQ ID NO: 2.
3. The polypeptide dimer according to claim 1, wherein the hinge of the first and second monomers is a hinge region derived from immunoglobulin IgD or a variant thereof.
4. The polypeptide dimer according to claim 3, wherein the hinge region derived from immunoglobulin IgD or a variant thereof contains at least one cysteine.
5. The polypeptide dimer according to claim 3, wherein the hinge region derived from immunoglobulin IgD or a variant thereof comprises the sequence of TABLE-US-00008 (SEQ ID NO: 17) Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1 Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro, wherein Xaa1 is Lys or Gly, and Xaa2 is Glu, Gly, or Ser; or TABLE-US-00009 (SEQ ID NO: 18) Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro, wherein Xaa3 is Lys or Gly, and Xaa4 is Glu, Gly, or Ser.
6. The polypeptide dimer according to claim 3, wherein the hinge region derived from immunoglobulin IgD or a variant thereof comprises any one amino acid sequence selected from the group consisting of SEQ ID

NO: 3, SEQ ID NO: 4, and SEQ ID NO: 19.

7. The polypeptide dimer according to claim 1, wherein the sialic acid is N-acetylneuraminic acid.

8. A composition comprising a polypeptide dimer according to claim 1.

9. The composition according to claim 8, wherein the composition is for subcutaneous injection.

10. The composition according to claim 8, which is a pharmaceutical composition or a food composition.

11. The composition according to claim 8, wherein the composition is a pharmaceutical composition in a form of a transdermal patch or topical patch.

12. The composition according to claim 8, wherein the Fc region of the first and second monomers comprises the amino acid sequence of SEQ ID NO: 2; and wherein the hinge is a hinge region derived from immunoglobulin IgD or a variant thereof, and wherein the sialic acid is N-acetylneuraminic acid.

13. The composition according to claim 8, wherein the hinge region derived from immunoglobulin IgD or a variant thereof comprises the sequence of TABLE-US-00010 (SEQ ID NO: 17) Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1 Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro, wherein Xaa1 is Lys or Gly, and Xaa2 is Glu, Gly, or Ser; or TABLE-US-00011 (SEQ ID NO: 18) Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro, wherein Xaa3 is Lys or Gly, and Xaa4 is Glu, Gly, or Ser.
